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SYSTEM 5.0: A Unified

Neuro-Semiotic Framework for

Symbolic Information Load, Limbic

Resonance, and Cultural Shock

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Abstract

We propose SYSTEM 5.0, a comprehensive, physically grounded framework unifying neuroaesthetics, predictive coding, and cultural evolution. We formalize artistic value across

three axes (Technique, Expressivity, Innovation) and four dimensions of verisimilitude (Biological, Spectral, Technical, Informational). Building on this, we introduce the theory of

informational telepathy: the capacity of the human limbic system to predict intentions via

the decoding of compressed symbolic information load. We model collective dynamics using

symbolic frequency and *limbic synchrony*, explaining phenomena ranging from orchestral

coordination to shared altered states of consciousness (e.g., dreams, near-death experiences, psychedelic visions) as instances of *limbic resonance*. Furthermore, we formalize the *symbolic decoding gap* to explain cultural shock in isolated populations. All mechanisms are falsifiable, with operational metrics provided.

Keywords: neuroaesthetics; informational telepathy; predictive coding; limbic resonance; symbolic decoding gap; altered states of consciousness

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1. Introduction

Formal theories of art, communication, and cultural evolution face a persistent tension between subjective phenomenological experience and measurable biological structure.

Classical accounts often reduce art to resemblance or technical skill, while computational approaches attempt to quantify aesthetic value through image statistics or

algorithmic

complexity. Concurrently, human communication is frequently described metaphorically as

"reading minds," yet no physical mechanism has been formally proposed to explain how

static symbols enable one individual to predict the future actions of another with high

accuracy without direct neural coupling.

SYSTEM 5.0 resolves these disparate threads into a unified neuro-semiotic framework. We

propose that aesthetic evaluation, interpersonal prediction, and collective cultural states are

all governed by the limbic system's capacity for predictive decoding of compressed symbolic

information.

1.1 Core Hypotheses

H₁ (Multidimensional Art & Absurdity): Artistic value is a three-dimensional vector

(Technique, Expressivity, Innovation) evaluated against four types of verisimilitude.

"Perfection" is an operational threshold that triggers a measurable neurocognitive response of absurdity.

H₂ (Informational Telepathy): Symbols carry compressed informational load. High-load symbols activate hippocampal-amygdalar-prefrontal circuits, generating

probabilistic behavioral models of the symbol's producer—a physical, biological form of telepathy.

H₃ (Cultural Shock & The Decoding Gap): Difficulties in cultural assimilation (e.g.,

isolated populations) are not cognitive deficits but predictive collapses caused by an exponentially widening symbolic decoding gap.

H₄ (Limbic Resonance & Altered States): Collective synchrony and shared altered

states (dreams, NDEs, psychedelics) are manifestations of limbic resonance, occurring

when external symbolic frequency drops, unmasking universal spectral archetypes.

2. The Neuro-Semiotic Foundations of Art and Verisimilitude

2.1 The Three Axes of Artistic Value

Let W denote an artwork or symbolic artifact. We represent its artistic profile as a vector:

$$A(W) = (T(W), E(W), I(W)) \in [0, 10]^3$$

where T is Technique (mastery of medium), E is Expressivity (capacity to evoke limbic affect), and I is Innovation (departure from established conventions).

2.2 The Four Dimensions of Verisimilitude

Verisimilitude is decomposed into four components operating on distinct temporal scales:

Biological (VB): Similarity to human morphology (Timescale: 500–1,000 years).

Spectral (VS): Activation of archetypal, culturally stabilized visual patterns (Timescale: 50–100 years).

Technical (VT): Fidelity to physical reality via technological mediation (Timescale: 3–8 years).

Informational (VI): The density of compressed symbolic data activating limbic predictions (Timescale: 1–5 years).

2.3 Perfection and the Neurocognitive Response of Absurdity

An artwork W achieves operational perfection if:

$$\min\{T, E, I\} \geq 9.5 \wedge \min\{VB, VS, VT, VI\} \geq 0.95 \wedge Q(W) \geq 9.0$$

where $Q(W)$ is a composite quality score (effort, iteration, peer feedback, originality).

Perfection triggers Absurdity (A), a measurable response combining cognitive surprise and limbic reward:

$$A(W) = (\Delta NAcc(W), \tau_{fix}(W), R_{verb}(W))$$

where $\Delta NAcc$ is the relative increase in nucleus accumbens activation (fMRI), τ_{fix} is visual fixation duration (eye-tracking), and R_{verb} is the subjective report of awe or impossibility.

3. Informational Telepathy: Limbic Prediction via Symbolic Load

3.1 Informational Load

We define the informational load $L(s)$ of a symbol s as the total number of emotional-predictive bits activated in the limbic system:

$$L(s) = L_{semantic}(s) + L_{emotional}(s) + L_{predictive}(s) + L_{contextual}(s)$$

While an isolated letter carries 0–10 bits, a modern technical symbol (e.g., a legal contract or

UI interface) carries 10^5 – 10^6 bits, requiring years of limbic calibration to

decode.

3.2 The Four-Node Limbic Decoding Circuit

Informational telepathy operates via a specific neural circuit:

Hippocampus (Context Retrieval): Generates a contextual prior $P(\text{context} | s)$.

Amygdala (Valence Assignment): Assigns emotional valence $v(s) \in [-1, 1]$ and

arousal $a(s) \in [0, 1]$.

Mirror Neuron System (Motor Simulation): Generates an embodied prediction of

implied actions.

Prefrontal Cortex (Integration): Outputs a coherent predictive model $\hat{a}$ of the producer's next action a .

The formal prediction model is:

$$\hat{a} = \arg \max_a P(a | s, \text{context}, v(s), a(s))$$

Telepathic Accuracy is defined as $1 - H(\hat{a} | a)$, where H is the conditional entropy between predicted and actual actions.

4. Collective Dynamics: Frequency, Synchrony, and Resonance

4.1 Symbolic Frequency and Limbic Synchrony

Let f_{sym} be the symbolic frequency (rate of introduction of new symbols into an ecosystem).

Let $\Sigma \in [0, 1]$ be limbic synchrony (the degree to which a group decodes symbols with identical predictive models).

Highly synchronized groups (e.g., orchestras, elite dance troupes) exhibit Informational

Telepathy in High Synchrony. The "sheet music" or "choreography" serves as a shared

symbolic alphabet. Musicians do not merely react; they predict the actions of their peers via

mirror-neuron simulation, achieving $\Sigma \rightarrow 1$.

4.2 Limbic Resonance in Altered States of Consciousness

SYSTEM 5.0 provides a non-mystical explanation for shared experiences in altered states

(dreams, near-death experiences [NDEs], and psychedelic visions).

Under normal waking consciousness, high external f_{sym} dominates limbic processing. In

altered states (e.g., via 5-HT_{2A} agonists like psilocybin, or hypoxia in NDEs), external f_{sym}

$\rightarrow 0$. The cortical filter is inhibited, and the limbic system defaults to

processing innate,
high-load Spectral Archetypes (e.g., tunnels, light, geometric entities, parental figures).

Because the human limbic hardware and spectral archetypes are universally shared,

individuals in altered states experience Limbic Resonance—they report identical "realms"

(e.g., heaven, hell, machine-elf encounters) not because these places exist externally, but

because the limbic system is decompressing the same universal biological archive in the

absence of external symbolic noise.

5. Cultural Evolution and Shock

5.1 The Symbolic Decoding Gap

Let SM be the modern symbolic ecosystem and SI be that of an isolated population. The

symbolic decoding gap is:

$$\Delta_{\text{sym}} = L(\text{SM}) - L(\text{SI} \cap \text{SM})$$

As technological complexity grows, $\Delta_{\text{sym}} \rightarrow \infty$. When an isolated individual encounters

modern symbols, the hippocampus retrieves no context ($P \approx 0$), and the amygdala defaults

to threat ($v < 0$, $a \rightarrow 1$). This results in Predictive Collapse—manifesting as cultural shock,

anxiety, and withdrawal. This is a decoding failure, not a cognitive deficit.

5.2 Informational Colonialism

Because modern individuals share basal biological/spectral verisimilitude with isolated

populations, they can partially predict isolated behaviors. The reverse is impossible. This

Asymmetric Telepathy maps directly onto colonial power structures, allowing dominant

groups to manipulate or marginalize isolated groups via indecipherable symbolic systems

(e.g., legal codes, bureaucratic interfaces).

6. Operational Metrics and Falsifiability

Table 1. Proposed Experimental Measurements

Concept Measurement Method Falsification Criterion

Informational Load $L(s)$ Free association latency + fMRI

BOLD signal volume

No correlation between $L(s)$ and

limbic activation volume

Telepathic Accuracy Subject predicts action from text;
compare to actual

Accuracy $\leq$ chance (25% for 4
options)

Symbolic Decoding Gap Time to predict action from
modern symbol (Isolated vs.
Control)

Gap not significantly larger than
within-group baseline

Limbic Resonance fMRI hyperscanning during altered
states / group rituals

Inter-subject correlation $\kappa \leq 0.2$

Absurdity Δ NAcc (fMRI) + τ fix (Eye-tracking) No significant spike in
reward/fixation metrics for
"perfect" art

7. Discussion and Ethical Considerations

7.1 Relation to Existing Literature

7.2 Ethical Implications

Informed Consent: Symbolic decoding and informational telepathy research
must

respect consent, especially when studying isolated populations.

Vulnerable Populations: Isolated individuals and those with anxiety disorders
are highly

susceptible to predictive collapse.

Democratic Erosion: State or corporate actors could deploy high-L(s)
artifacts to

replace democratic deliberation with amygdalar reflex.

7.3 Limitations

8. Conclusion

Acknowledgments

Conflicts of Interest

Data Availability

References

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature
Reviews*

Neuroscience, 11(2), 127–138.

Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the
future of

cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.

Rizzolatti, G., & Craighero, L. (2004). The mirror-neuron system. *Annual Review
of*

Neuroscience, 27, 169–192.

Shannon, C. E. (1948). A mathematical theory of communication. Bell System Technical Journal, 27(3), 379–423.

Barrett, L. F. (2017). How Emotions Are Made: The Secret Life of the Brain. Houghton Mifflin Harcourt.

Chatterjee, A., & Vartanian, O. (2014). Neuroaesthetics. Trends in Cognitive Sciences, 18(7), 370–375.

LeDoux, J. E. (2000). Emotion circuits in the brain. Annual Review of Neuroscience, 23, 155–184.

Carhart-Harris, R. L., et al. (2012). Neural correlates of the psychedelic state as determined by fMRI studies with psilocybin. PNAS, 109(6), 2138-2143.

Boyd, R., & Richerson, P. J. (1985). Culture and the Evolutionary Process. University of Chicago Press.

Simon, H. A. (1971). Designing organizations for an information-rich world. In Computers, Communications, and the Public Interest.

Dehaene, S. (2009). Reading in the Brain: The New Science of How We Read. Viking.

Henrich, J. (2015). The Secret of Our Success: How Culture Is Driving Human Evolution. Princeton University Press.

Appendix A: Formal Notation Summary

Symbol Meaning

W Artwork / Symbolic Artifact

T, E, I Technique, Expressivity, Innovation scores

VB, VS, VT, VI Biological, Spectral, Technical, Informational Verisimilitude

Q(W) Composite Quality score

A(W) Absurdity response vector

s A symbol (letter, word, image, icon)

L(s) Informational load of s (emotional-predictive bits)

$\hat{a}$, a Predicted action, Actual action

fsym Symbolic frequency (rate of ecosystem update)

Σ Limbic synchrony (inter-subject correlation)

SM, SI Modern / Isolated symbolic ecosystems

Δ sym Symbolic decoding gap

End of manuscript.

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